

Equal Entropy Spacing Selects the 8×8 Torus: A Geometric Derivation of $d = 4$ from the Laplacian Spectrum

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Warped Information Number (WIN) Paradigm - White Paper

Status: Research finding - computational derivation, no fitted parameters

Abstract

The WIN Paradigm derives the Standard Model from the spacetime dimension $d = 4$. Prior work took $d = 4$ as an input derived from the photon's helicity structure, or from the requirement that the $L \times L$ torus host exactly 14 modes at its midpoint. Both anchors are interpretive: the first maps photon phase space onto dimension, the second uses the Standard Model particle count as a premise.

This paper presents a geometric selector that does not use the particle count as input. The $L \times L$ torus Laplacian produces standing wave modes whose Shannon entropy takes a discrete set of values the entropy classes of the torus. For each L , the number of classes and their spacing are determined by the divisor structure of L . We show:

1. The 8×8 torus is the unique $L \times L$ torus (for $4 \leq L \leq 10,000$) whose entropy spectrum has exactly three equally-spaced classes: 6.0, 5.5, 5.0 bits, spacing 0.5.
2. Among the 929 values of L in the tested range with three entropy classes, exactly one has equal spacing. There are no near misses: the next smallest spacing difference exceeds 10^{-2} , well above numerical precision. The property is robust.

The Standard Model organizes its fermions into three mass tiers (light, medium, heavy). If the substrate's entropy spectrum is what generates the tiers, the substrate must have three equally-spaced entropy classes. This selects $L = 8$, hence $N = L^2 = 64$, hence $d = 4$ via the framework relation $N = d \cdot 2^d$. The 14-mode multiplet is now a consequence, not a premise: $2(L - 1) = 14$ drops out once $L = 8$ is fixed.

This is the strongest anchor the framework has. It uses one physical input “the SM has three mass tiers” and one mathematical fact “only $L = 8$ produces three equally-spaced entropy classes” — to derive $d = 4$ and the particle count.

1 Introduction

1.1 The $d = 4$ Problem

The WIN Paradigm is a dimensional reduction framework: from $d = 4$, all Standard Model structure follows by arithmetic. The framework derives $N = d \cdot 2^d = 64$, the 8×8 torus, the 14-mode multiplet at $\lambda = 4$, the support classes $64, 48, 32 = N, V, N/2$, the δ function that assigns the 14 $N/4$ values, the mass map $m = A \exp(-B \cdot N/4)$, the warp factor $k_L = 38.44$, the Higgs mass 125.138 GeV, and the Weinberg angle 0.23135.

What the framework has not derived from first principles is $d = 4$ itself. Earlier anchors:

Anchor	Uses SM count?	Type
Photon has 2 helicities, 2 real components $\Rightarrow$ 4D phase space	No	Interpretive
$2(L - 1) = 14 \Rightarrow L = 8$	Yes (14)	Counting
$p(4) = 5 = \text{rank}(\text{SO}(10))$	No	Combinatorial

None is a derivation. The photon argument is a physical mapping, not a mathematical one. The counting argument starts from the particle count. The combinatorial argument uses the rank of $\text{SO}(10)$, which is itself a framework choice.

1.2 What This Paper Adds

A geometric selector rooted in the entropy spectrum of the torus, not in the particle count and not in a physical mapping. The result:

3 mass tiers $\Rightarrow$ 3 equally-spaced entropy classes $\Rightarrow L = 8 \Rightarrow N = 64 \Rightarrow d = 4 \Rightarrow 14$ modes

The SM count 14 is at the end of the chain, derived. It is not a premise.

2 Setup

2.1 The $L \times L$ Torus and its Laplacian

Let $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ be the $L \times L$ periodic square lattice. The discrete nearest-neighbour Laplacian has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{2\pi n_1}{L}\right) - 2 \cos\left(\frac{2\pi n_2}{L}\right), \quad (n_1, n_2) \in \{0, \dots, L-1\}^2. \quad (1)$$

The spectrum is mirror symmetric under $\lambda \rightarrow 8 - \lambda$ with equal multiplicity for every pair $(\lambda, 8 - \lambda)$, except $\lambda = 4$ which is the fixed point. At $\lambda = 4$, the multiplicity is exactly $2(L - 1)$.

2.2 Standing Waves and Shannon Entropy

The real spatial standing waves are

$$v(x, y) = \cos\left(\frac{2\pi(n_1 x + n_2 y)}{L}\right), \quad (x, y) \in \Lambda_L. \quad (2)$$

The Shannon entropy of a mode is

$$S(n_1, n_2) = - \sum_{(x, y) \in \Lambda_L} p(x, y) \log_2 p(x, y), \quad p(x, y) = \frac{|v(x, y)|^2}{\sum_{\Lambda_L} |v(x, y)|^2}. \quad (3)$$

The entropy is invariant under the D_4 torus symmetry. It depends only on the order of the mode vector, i.e., on

$$m = \frac{L}{\text{gcd}(n_1, n_2, L)}. \quad (4)$$

The integer m divides L . The number of distinct entropy values the entropy classes of L — equals the number of distinct values of $S(m, L)$ over the divisors m of L .

2.3 Analytic Formula for the Entropy Classes

The entropy of a mode with parameter m is

$$S(m, L) = \begin{cases} 2 \log_2 L & m \in \{1, 2\}, \\ 2 \log_2 L - 1 - \frac{2}{m} C_2(m) & m \geq 3, \end{cases} \quad (5)$$

where

$$C_2(m) = \sum_{k=0}^{m-1} \cos^2 \left(\frac{2\pi k}{m} \right) \log_2 \cos^2 \left(\frac{2\pi k}{m} \right). \quad (6)$$

This formula is exact. It is verified against direct enumeration for $L \in \{4, 6, 8, 12\}$ (Section 4.1). The entropy classes of L are then $C_L = \{S(m, L) : m \mid L\}$.

2.4 The Three Class Equal-Spacing Condition

We say L satisfies the equal spacing condition if $|C_L| = 3$ and $\Delta_1 = \Delta_2$, where $\Delta_1 = c_1 - c_2$, $\Delta_2 = c_2 - c_3$ are the spacings between the three classes (ordered descending).

3 The Result

3.1 Statement

Theorem (computational). *Among all integers L with $4 \leq L \leq 10,000$, the only L for which the $L \times L$ torus entropy spectrum has exactly three equally-spaced classes is $L = 8$.*

At $L = 8$, the three classes are $C_8 = \{6.0, 5.5, 5.0\}$, with spacing $\Delta_1 = \Delta_2 = 0.5$ bits.

3.2 Uniqueness and Robustness

The scan from $L = 4$ to $L = 10,000$ gives:

Quantity	Value
L with exactly three entropy classes	929
L with three equally spaced classes	1 (That $L = 8$)
Spacing difference at $L = 8$	0.0
Minimum spacing difference among all other three-class L	$> 10^{-2}$
Near-misses (difference $< 10^{-2}$ but nonzero)	0

The result is not marginal. The spacing difference at $L = 8$ is exactly zero (to machine precision), and no other L comes within 10^{-2} of the equal-spacing condition. There is no ambiguity, no tolerance tuning, no near call.

3.3 Structure of the Three-Class L Values

The 929 values of L with exactly three classes fall into recognizable families:

Family	Count	Example
p^2 for prime p	24	9, 25, 49, 121, ...
$2p^2$ for prime p	19	18, 50, 98, 242, ...
Other	886	1929, 2127, 2199, ...

Every one of these families has unequal spacing. The equality condition is satisfied only by $L = 8 = 2^3$, which is $2p^2$ with $p = 2$.

4 Verification

4.1 Sanity Check

The analytic formula is checked against direct enumeration:

L	Direct	Analytic	Match
4	{4.0, 3.0}	{4.0, 3.0}	✓
6	{5.1699, 4.8366}	{5.1699, 4.8366}	✓
8	{6.0, 5.5, 5.0}	{6.0, 5.5, 5.0}	✓
12	{7.1699, 6.8366, 6.7108, 6.1699}	Same	✓

The formula is exact.

4.2 Full Scan

Every L from 4 to 10,000 is checked for the equal-spacing condition. The result is unique: only $L = 8$.

4.3 Distance to the Condition

For every three-class L , the spacing difference $|\Delta_1 - \Delta_2|$ is computed. The minimum across all $L \neq 8$ is $> 10^{-2}$. The value at $L = 8$ is 0. No tolerance is required to distinguish.

5 Derivation Chain

The result fits into the framework’s existing chain as a new anchor:

$$\begin{array}{c}
\text{Standard Model has 3 mass tiers} \\
\Downarrow \text{ (physical identification)} \\
\text{Substrate must have 3 equally-spaced entropy classes} \\
\Downarrow \text{ (Section 3)} \\
L = 8 \\
\Downarrow \\
N = L^2 = 64 \\
\Downarrow \\
d = 4 \text{ (via } N = d \cdot 2^d) \\
\Downarrow \\
2(L - 1) = 14 \text{ modes at } \lambda = 4 \\
\Downarrow \\
\text{Standard Model particle multiplet}
\end{array}$$

The particle count 14 appears at the bottom of the chain. It is no longer an input.

6 What This Changes in the Framework

6.1 The 14-Mode Count is Derived

Prior papers derived $L = 8$ from the SM count 14 by the formula $2(L - 1) = 14$. This is circular at the level of the anchor: the SM count justifies the torus size, and the torus size hosts the SM count. The new chain does not use 14. It uses only the physical fact “3 mass tiers” and the mathematical fact “only $L = 8$ gives equally-spaced entropy classes.” The number 14 emerges as a consequence at the end.

6.2 Replacing the Photon Helicity Anchor

The photon-helicity anchor stated: photon has 2 helicities, each with 2 real components, so the phase space is 4D, so $d = 4$. This is an interpretive mapping. The entropy anchor is geometric and computational:

- It uses only the Laplacian spectrum of the torus.
- It is verified numerically to $L = 10,000$.
- It has no near-misses.

The entropy anchor is not a replacement for physical intuition — the “3 mass tiers” identification is still physical. But it is the strongest anchor the framework has, because it produces a unique, robust, geometric selector.

6.3 Relation to Prior Anchors

Anchor	Uses SM count?	Unique?	Type
Photon helicity	No	Yes (trivially)	Interpretive
$2(L - 1) = 14$	Yes	Yes	Counting
$p(4) = 5$	No	Yes	Combinatorial
Equal entropy spacing	No	Yes	Geometric

The entropy anchor is the only one that is both geometric and does not use the particle count as a premise.

7 What Is Derived and What Is Assumed

7.1 Derived (computational)

- The entropy class formula $S(m, L)$, verified against direct enumeration.
- The classification of L by number of entropy classes, for $4 \leq L \leq 10,000$.
- The uniqueness of $L = 8$ among all L with three equally-spaced classes.
- The exact spacing values 0.5, 0.5 at $L = 8$.
- The absence of near-misses (minimum spacing difference among other three-class L exceeds 10^{-2}).
- The family structure: p^2 , $2p^2$, and other three-class L all have unequal spacing.

7.2 Derived (framework chain)

- $N = 64$ from $L = 8$.
- $d = 4$ from $N = d \cdot 2^d$.
- $2(L - 1) = 14$ modes at $\lambda = 4$.
- The support classes $64, 48, 32 = N, V, N/2$.
- The full particle spectrum, given the framework’s other derivations.

7.3 Assumed

- **The Standard Model has three mass tiers.** This is a physical fact about the fermion spectrum.
- **The substrate’s entropy spectrum generates the tiers.** This is the physical identification that connects the mathematical fact to the physical world.
- $N = d \cdot 2^d$. The framework relation connecting lattice size to spacetime dimension. Derived elsewhere in the framework, but not in this paper.

7.4 Range of Verification

The uniqueness of $L = 8$ is verified for $4 \leq L \leq 10,000$. It is not proven for all L . The pattern strongly suggests it holds generally — the entropy spacing is a number-theoretic function of the divisor structure of L , and for $L > 10^4$ the divisor structure becomes sparser, not denser, making equal spacing even less likely — but the proof is a separate task.

8 Open Problems

1. **Analytic proof of uniqueness.** Show that $L = 8$ is the unique solution to the equation $\Delta_1(L) = \Delta_2(L)$ for all $L \geq 4$. The entropy formula $S(m, L)$ involves the sums $C_2(m)$, which are Bessel-like. A closed-form inequality relating $C_2(m)$ for divisors $m \mid L$ would close the proof.
2. **Extension to $L > 10^4$.** Verify computationally at $L = 10^5$ and 10^6 if resources permit. The pattern is expected to hold.
3. **Interpretation of “3 mass tiers.”** The identification of the SM’s three tiers (light, medium, heavy) with three entropy classes is a physical identification. Is it derivable from the SM’s gauge structure, or is it an independent fact about the fermion spectrum?
4. **Relation to the $2(L-1) = 14$ anchor.** The two derivations of $L = 8$ (particle-count and entropy) should be shown to agree exactly. Do they give the same L for all combinations of inputs, or is the entropy anchor strictly stronger?
5. **The $N = d \cdot 2^d$ relation.** The derivation of $d = 4$ from $N = 64$ requires this relation. Its justification is outside the scope of this paper, but the reader should see where the chain enters the wider framework.

9 Conclusion

The 8×8 torus is the unique $L \times L$ torus (for $4 \leq L \leq 10,000$) whose Laplacian entropy spectrum has exactly three equally-spaced classes: 6.0, 5.5, 5.0 bits, spacing 0.5. Among 929 three-class L values, exactly one has equal spacing, and the next-closest spacing difference exceeds 10^{-2} — no near-misses.

The Standard Model organizes its fermions into three mass tiers. If the substrate’s entropy spectrum generates the tiers, the substrate must have three equally spaced entropy classes. This selects $L = 8$, hence $N = 64$, hence $d = 4$ via $N = d \cdot 2^d$. The 14-mode multiplet at $\lambda = 4$ is a consequence of $L = 8$: $2(L - 1) = 14$.

This is the framework’s strongest anchor for $d = 4$. It is geometric, computationally verified, robust, and does not use the Standard Model particle count as input. The particle count is derived, not assumed. The 8×8 torus is selected by entropy, not by counting.

A Numerical Values

Quantity	Value
d	4
N	64
L	8
Entropy classes at $L = 8$	6.0, 5.5, 5.0 bits
Spacing	0.5, 0.5 bits
L with 3 classes, $4 \leq L \leq 10^4$	929
L with 3 equally-spaced classes	1
Unique L	8
Near-misses ($\Delta_1 - \Delta_2 < 10^{-2}$)	0
Minimum spacing difference among other 3-class L	$> 10^{-2}$
p^2 family count	24
$2p^2$ family count	19

References

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